



Genotoxicity of mixtures of glyphosate and atrazine and their environmental transformation products before and after photoactivation



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HIGHLIGHTS

- The cytogenetic effects of pesticide mixtures were evaluated on CHO cells.
- A 20-fold enhanced activity was observed in mixture with the four pesticides.
- It was 100-fold increased after light-irradiation, through oxidative stress.
- It highlighted the importance of cocktail effects in environmental matrices.
- It showed the limits of usual strategies to estimate environmental risks.

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ABSTRACT

The photo-inducible cytogenetic toxicity of glyphosate, atrazine, aminomethyl phosphoric acid (AMPA), desethyl-atrazine (DEA), and their various mixtures was assessed by the in vitro micronucleus assay on CHO-K1 cells.

Results demonstrated that the cytogenetic potentials of pesticides greatly depended on their physico-chemical environment. The mixture made with the four pesticides exhibited the most potent cytogenetic toxicity, which was 20-fold higher than those of the most active compound AMPA, and 100-fold increased after light-irradiation. Intracellular ROS assessment suggested the involvement of oxidative stress in the genotoxic impact of pesticides and pesticide mixtures.

This study established that enhanced cytogenetic activities could be observed in pesticide mixtures containing glyphosate, atrazine, and their degradation products AMPA and DEA. It highlighted the importance of cocktail effects in environmental matrices, and pointed out the limits of usual testing strategies based on individual molecules, to efficiently estimate environmental risks.

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1. Introduction

Since their discovery in the 1940s, pesticides have greatly contributed to improve the yield and quality of plants, and to ensure crop production. As a consequence, the global consumption of pesticides, which was lower than 0.5 kg Ha⁻¹ in the 1960s, has constantly increased to reach 2 Kg Ha⁻¹ in 2004. Today, modern agriculture is very dependent on pesticides, since 4.6 million tons of chemical pesticides are annually sprayed into the environment, corresponding to an approximate global sale of 300 billion dollars (Zhang et al., 2011). Despite various international regulations,

pesticides have contaminated all the environmental compartments worldwide (Vecchia et al., 2009; Dubois and Lacouture, 2011; Mouvet, 2007). In France, 35% of the soil surface is subjected to pesticide contamination, and fifteen pesticides are commonly quantified in soils and waters. They mostly correspond to substances which have been banned since the 2000s, or to their degradation products and metabolites (Dubois and Lacouture, 2011).

Atrazine and glyphosate have been among the most widely used agricultural pesticides (Zeljezic et al., 2006; Mañas et al., 2009a). With nearly 650 000 tons annually widespread, glyphosate is the best-selling chemical pesticide in the world. It is also the top ranked herbicide in the European Union, as it has been used in 40% of agricultural land (Zhang et al., 2011). Atrazine is the most popular triazine pesticide worldwide (Pathak and Dikshit, 2012). In the early 90s, it was spread on approximately 67% of all corn acreage, 65% of sorghum acreage, and 90% of sugar-cane acreage. Today,

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glyphosate, atrazine and their main breakdown products aminomethyl phosphoric acid (AMPA) and desethyl-atrazine (DEA), are responsible for an almost universal contamination of surface water and groundwater (Dubois and Lacouture, 2011).

Because of their intrinsic toxicity and their persistent nature, pesticides have greatly contributed to degrade ecosystems (Joy et al., 2005). As a consequence, their residual presence in plants and animals has resulted in a global contamination of food, which represents an important health risk to human populations. Indeed, occupational exposure to pesticides has led to a significantly increased risk of degenerative diseases such as neoplasia (Bolognesi et al., 2009; Barry et al., 2011) and neuro-endocrine disruptions (Cooper et al., 2007). Moreover, environmental exposure to pesticides has been suspected to generate long-term transmissible effects, since recent studies have demonstrated that some of them could interfere with DNA metabolism (Mineau, 2005). Concerning glyphosate and atrazine, several experiments have attempted to assess and quantify their genotoxic/mutagenic properties. In spite of conflicting results mostly due to the different methodologies used, some studies have established their genotoxic potential. Glyphosate has been shown to induce in vitro increases of sister chromatid exchanges, chromosome aberrations (Lioi et al., 1998a; Lioi et al., 1998b) and DNA lesions in human cells (Monroy et al., 2005; Mladinic et al., 2009; Mañas et al., 2009a). It has also produced in vivo genotoxic and clastogenic damages (Prasad et al., 2009; Mañas et al., 2009a). Atrazine has been classified as “possibly carcinogenic to human” (Group 2B) by the International Agency for Research on Cancer (1999). It has been shown to induce both in vitro and in vivo cytogenetic damages (Biradar and Rayburn, 1995; Gebel et al., 1997). On the contrary, very few studies have been conducted on their main degradation products AMPA and DEA. Moreover, most of the assays have been performed in laboratory conditions, which considerably differed from those observed in the environment. In particular, the possible photo-inducible properties of such compounds have certainly been underestimated, as well as the possible enhanced effects of pesticide mixtures.

In the present study, the in vitro micronucleus assay was used to evaluate the cytogenetic toxicity of glyphosate and atrazine, and their degradation products AMPA and DEA. Then, to mimic the environmental conditions that may occur in soils or waters, different pesticide mixtures were prepared with glyphosate, atrazine, AMPA and DEA, and assessed for their cytogenetic toxicity. For each pesticide or pesticide mixture, assays were performed in the dark with and without metabolic activation, and with sunlight irradiation. Intracellular ROS amounts were also analyzed in the dark to evaluate oxidative stress.

2. Materials and methods

2.1. Reagents and cell culture

Atrazine ($C_8H_{14}ClN_5$, CAS No. 1912-24-9), glyphosate (N-(phosphonomethyl)glycine, $C_3H_8NO_5P$, CAS No. 1071-83-6), DEA (desethyl-atrazine, 2-Amino-4-chloro-6-isopropylamino-1,3,5-triazine, $C_6H_{10}ClN_5$, CAS No. 6190-65-4) and AMPA (Aminomethyl phosphonic acid, $NH_2CH_2P(O)(OH)_2$, CAS No. 1066-51-9) were purchased from Sigma–Aldrich Chemical Company (St Quentin-Fallavier, France). Atrazine and DEA were dissolved into dimethyl sulfoxide DMSO (Sigma) while glyphosate and AMPA were dissolved in phosphate buffer saline PBS (Sigma).

To fully explore the combinatory effects of pesticides, all the possible combinations of the four pesticides were assessed. Six different mixtures of two pesticides were made: glyphosate/AMPA, glyphosate/atrazine, glyphosate/DEA, AMPA/atrazine, AMPA/DEA

and atrazine/DEA. For each mixture, a stock solution containing 10 mg mL^{-1} of the sum of the different compounds (5 mg mL^{-1} of each pesticide) was obtained, and several dilutions of this mixture were performed. Four different mixtures of three pesticides were made: glyphosate/AMPA/atrazine, glyphosate/AMPA/DEA, glyphosate/atrazine/DEA and AMPA/atrazine/DEA. For each mixture, a stock solution containing 10 mg mL^{-1} of the sum of the different compounds (3.33 mg mL^{-1} of each pesticide) was obtained, and several dilutions of this mixture were performed. A mixture was made with the four pesticides: glyphosate/AMPA/atrazine/DEA. A stock solution containing 10 mg mL^{-1} of the sum of the different compounds (2.5 mg mL^{-1} of each pesticide) was obtained, and several dilutions of this mixture were performed.

Experiments were performed on the well-established Chinese Hamster Ovary cell line, CHO-K1 cells (ATCC-LGC Standards Sarl, Molsheim, France). This cell line is characterized by a good genetic stability and by a short generation time, it has been well validated for genetic toxicology assays (Westerink et al., 2011). CHO-K1 cells were maintained in McCoy's 5A medium (Sigma, St Quentin-Fallavier, France) supplemented with 10% bovine calf serum, 1 mM glutamine (1 mM, Eurobio, Paris, France), and 100 U mL^{-1} – $10\text{ }\mu\text{g mL}^{-1}$ penicillin–streptomycin (Eurobio, Paris, France). They were incubated at $37\text{ }^\circ\text{C}$ in a humidified atmosphere containing 5% CO_2 .

2.2. Metabolic activation mixture (S9 Mix)

To examine the possible clastogenic effects of pesticide metabolites resulting from the cytochrome-based P450 oxidation systems in mammalian livers, an exogenous activation system (S9 Mix) was used. This metabolic activation system (S9) was a 9000-g centrifuged supernatant of a 10% liver homogenate and was prepared from male Sprague–Dawley rats treated with a single injection of Aroclor 1254 (500 mg kg^{-1} body weight), 5 d before they were killed. The protein concentration in the S9 homogenate was 26 mg mL^{-1} . In the micronucleus assay, S9 Mix contained 10% S9, 5 mM glucose-6-phosphate, 4 mM NADP, 33 mM KCl and 8 mM MgCl_2 diluted in PBS.

2.3. Photoactivation

To reproduce the photo-inducible phenomenon experienced by exposition to pesticides in the environment, the micronucleus assay was performed with a photoactivation procedure. Irradiation of cell cultures was carried out 1 h after the addition of pesticides. Irradiation was performed with a solar simulator Suntest CPS+ (Atlas Material Testing Technology BV, Mousy le Neuf, France) apparatus equipped with a xenon arc lamp (1100 W), a special glass filter restricting transmission of light below 290 nm and a near IR-blocking filter. The irradiance for the photoactivation was fixed at 750 W m^{-2} throughout the experiments and the combined light dose was 4.5 J cm^{-2} for one minute irradiation (0.03 J cm^{-2} of UVB, 0.41 J cm^{-2} of UVA and 4.06 J cm^{-2} of visible light). This irradiation dose corresponded to 1–3 min period of solar exposure during a clear summer day in the United Kingdom (Diffey, 2002).

The temperature of the samples was kept at $4\text{ }^\circ\text{C}$ using a water cooling circuit in the irradiation chamber. UVA-visible light (320–800 nm) was obtained using the solar ID65 filter plus a window glass filter.

2.4. Micronucleus assay

Micronuclei are defined as chromosome fragments or whole chromosomes that lag during cell division due to the lack of a centromere or to a defect in cytokinesis (Kirsch-Volders et al.,

1997). They may be produced by clastogenic or aneugenic compounds, according to a wide range of mechanisms that include both genotoxic and epigenetic events. The micronucleus assay allows the scoring of micronuclei in the cytoplasm of interphasic cells exposed in vitro or in vivo to clastogenic and/or aneugenic agents.

The capacity of pesticides to induce micronuclei was assessed according to the protocol described by Kirsch-Volders et al. (1997). A total of 50000 CHO-K1 cells was plated in chamber slides and incubated for 24 h at 37 °C in a humidified atmosphere containing 5% CO₂. Various concentrations of pesticides and pesticide mixtures (from 10⁻⁴ to 100 µg mL⁻¹) were incorporated into duplicate CHO-K1 cell cultures. Three negative controls were added to determine the background DNA-damage levels in CHO cells in the dark and with light irradiation: the culture medium only, DMSO and PBS. Two positive controls with well-known genotoxic compounds were also included to ensure the sensitivity of the assay: these were 0.06 µg mL⁻¹ mitomycin C without S9 Mix and 5 µg mL⁻¹ benzo[a]pyrene (BaP) with S9 Mix.

After a 3-h exposure, cells were rinsed with PBS and incubated in fresh medium containing cytochalasin B (3 µg mL⁻¹) for an additional 24 h to stop cytokinesis. At the end of the incubation period, cells were submitted to two successive washes with PBS and fixed with methanol (HPLC purity grade solvent). The slides were air dried and stained with 5% Giemsa stain in Milli-Q water for 15 min.

The Cytokinesis Blocked Proliferation Index (CBPI) was used to select adequate concentrations for the assessment of micronuclei, as it has been considered a measure of cytotoxicity (Kirsch-Volders et al., 1997). CBPI was determined by scoring the number of mononucleated (M1), binucleated (M2), and trinucleated (M3) cells among 500 Giemsa-stained cells with well-preserved cytoplasm: $CBPI = (M1 + 2 \cdot M2 + 3 \cdot M3) / 500$. When a cytotoxic effect was observed, micronucleated cell rates were determined for concentrations inducing less than a 50% decrease of CBPI. When no cytotoxic effect was observed, the maximal concentration assessed was 100 µg mL⁻¹. When a clastogenic/aneugenic activity was observed, four relevant concentrations were selected to obtain a dose–response relationship.

A total of 2000 binucleated cells were examined for each concentration, and micronuclei were identified according to the morphological criteria previously defined by Kirsch-Volders et al. (1997). Statistical differences between negative controls and treated samples were determined using the ² test. The standard micronucleus assay was considered positive when a dose–response relationship could be established between the numbers of micronucleated cells and the concentrations of pesticide solutions, and when at least one concentration induced a significant increase of micronucleated cells as compared to the medium-only control culture. The dose–response relationships were calculated by non linear regression analysis with TableCurve2D®. Model significance was based on three criteria: (i) correlation coefficient r^2 being > 0.62; (ii) model probability (P) being < 0.05; (iii) error probability (P_e) being > 0.05. The Minimal Clastogenic Concentration (MCC) was defined as the lowest concentration of pesticide (µg mL⁻¹) that induced a significant increase of micronucleated cells. The Cytogenetic Potency (CP) was defined as the slope of the dose–response curves. It was calculated by non-linear regression analysis with TableCurve2D®.

2.5. Intracellular ROS analysis

Oxidative stress was evaluated by assessing intracellular ROS amounts, using the OxiSelect™ Intracellular ROS Assay Kit (Cell Biolabs, Paris, France). It is a cell-based assay which employs the 2',7'-Dichlorodihydrofluorescein diacetate (DCFH-DA). After diffusion into the cell, this fluorescent probe is deacetylated by cellular esterases to non-fluorescent 2',7'-Dichlorodihydrofluorescein

(DCFH), which is rapidly oxidized to highly fluorescent 2', 7'-Dichlorodihydrofluorescein (DCF) by ROS. A total of 50000 cells were seeded in a black 96-well cell culture plate and incubated at 37 °C for 24 h. Cells were rinsed 3 times with PBS and incubated into 100 µL of DCFH-DA/media solution at 37 °C for 60 min. At the end of the incubation period, cells were rinsed with PBS and treated with pesticides or pesticide mixtures. Experiments were performed in triplicate. Four concentrations were tested for each pesticide or pesticide mixture; these concentrations were the same as those assessed for the micronucleus assay. H₂O₂ (100 µM) was used as positive control. After a 30-min incubation period, fluorescence was measured with a fluorometric plate reader at 480 nm/530 nm. The dose–response relationships were calculated by linear regression analysis with TableCurve2D®. Model significance was based on three criteria: (i) correlation coefficient r^2 being > 0.62; (ii) model probability (P) being < 0.05; (iii) error probability (P_e) being > 0.05. The Oxidative Potency was defined as the slope of the dose–response curves; it was calculated according to a standard curve obtained with various concentrations of fluorescent DCF.

3. Results

The cytogenetic toxicities of glyphosate, AMPA, atrazine and DEA are summarized in Table 1, and reported in Fig. 1. AMPA displayed a strong photo-inducible clastogenic/aneugenic activity (MCC = 0.006 µg mL⁻¹ in the dark and MCC = 0.0004 µg mL⁻¹ after light-irradiation). Atrazine and glyphosate displayed cytogenetic activity after metabolic activation (MCC = 0.064 µg mL⁻¹ and MCC = 5.8 µg mL⁻¹ respectively). DEA was inactive in all the experimental conditions.

The cytogenetic toxicities observed for combinations of two pesticides are reported in Table 2 and in Fig. 1. Surprisingly, they were globally lower than those obtained with the most active individual compounds AMPA and atrazine, suggesting the absence of cocktail effect. Indeed, only atrazine/AMPA (MCC = 0.39 µg mL⁻¹ in the dark and MCC = 0.0026 µg mL⁻¹ after light irradiation) and glyphosate/DEA (MCC = 22.1 µg mL⁻¹ after metabolic activation), displayed significant but moderated effects.

The cytogenetic toxicities observed for combinations of three and four pesticides are reported in Table 3 and Fig. 1. A strong cytogenetic effect was observed for the glyphosate/atrazine/AMPA mixture in the dark (MCC = 0.001 µg mL⁻¹), and all the pesticide combinations except the atrazine/AMPA/DEA mixture were activated by light-irradiation. However, their cytogenetic potentials were close to that of AMPA, indicating weak synergistic effects. The mixture made with the four pesticides, on the contrary, exhibited a very powerful cytogenetic activity, with MCC lower than 0.001 µg mL⁻¹ in all the experimental conditions. This activity was 20-fold higher than that of AMPA in the dark (MCC = 0.0004 µg mL⁻¹), and 200 fold higher than that of atrazine after metabolic activation (MCC = 0.0003 µg mL⁻¹). It was also strongly photo-inducible, since it was 100 fold increased by light-irradiation (MCC = 4.10⁻⁶ µg mL⁻¹).

Oxidative stress induced by pesticides and pesticide mixtures are reported in Fig. 2. The oxidative potency of H₂O₂, used as positive control, was 333.3 nM DCF/µM⁻¹ (data not shown). Only AMPA exerted an elevated oxidative effect (5.9 nM DCF/µg mL⁻¹), whereas the oxidative potencies of glyphosate, atrazine and DEA were very low. Among pesticide mixtures, the atrazine/AMPA and glyphosate/atrazine/AMPA combinations displayed high oxidative properties, with oxidative potencies reaching 7.8 and 57 nM DCF/µg mL⁻¹ respectively. The mixture made with the four pesticides exhibited the strongest oxidative potency (1125 nM DCF/µg mL⁻¹).

Table 1
Summary data concerning the cytogenetic effects of pesticides.

	In the dark				With S9 mix				With light-irradiation			
	Doses ($\mu\text{g mL}^{-1}$)	CBPI (%)	BMC (%)	MCC ($\mu\text{g mL}^{-1}$)	Doses ($\mu\text{g mL}^{-1}$)	CBPI (%)	BMC (%)	MCC ($\mu\text{g mL}^{-1}$)	Doses ($\mu\text{g mL}^{-1}$)	CBPI (%)	BMC (%)	MCC ($\mu\text{g mL}^{-1}$)
Negative control		100	9.5 $\pm$ 0.7		100	9.5 $\pm$ 0.7			100	11.5 $\pm$ 0.7		
Positive control		99	30 $\pm$ 2.8***		98	22 $\pm$ 2.8***			99	36 $\pm$ 2.8***		
Glyphosate	5	96	9 $\pm$ 2.8	–	5	98	17 $\pm$ 1.4	5.8	5	100	9.5 $\pm$ 0.7	93.4
	10	90	8.5 $\pm$ 0.7		10	92	39 $\pm$ 2.8***		10	95	10.5 $\pm$ 0.7	
	50	88	9.5 $\pm$ 0.7		50	89	41 $\pm$ 4.2***		50	89	16.5 $\pm$ 2.8	
	100	84	13 $\pm$ 1.4		100	85	42.5 $\pm$ 2.8***		100	78	20.5 $\pm$ 2.1*	
Atrazine	5	99	8.5 $\pm$ 0.7	–	0.05	99	17 $\pm$ 1.4	0.064	5	100	12 $\pm$ 1.4	–
	10	98	10 $\pm$ 1.4		0.1	96	21 $\pm$ 1.4**		10	98	11 $\pm$ 1.4	
	50	96	9 $\pm$ 1.4		0.5	94	26 $\pm$ 2.8***		50	96	10.5 $\pm$ 0.7	
	100	94	11.5 $\pm$ 0.7		1	92	34 $\pm$ 2.8***		100	91	13 $\pm$ 1.4	
AMPA	0.005	100	17 $\pm$ 1.4	0.006	0.1	97	12.5 $\pm$ 2.1	0.78	0.00005	100	11.5 $\pm$ 3.5	0.0004
	0.01	99	20 $\pm$ 2.8*		0.5	94	17 $\pm$ 2.8		0.0001	100	16.5 $\pm$ 2.1	
	0.05	98	21 $\pm$ 4**		1	90	21 $\pm$ 1.4**		0.0005	99	24 $\pm$ 2.8**	
	0.1	95	23 $\pm$ 1.4***		5	86	25 $\pm$ 1.4***		0.001	98	30 $\pm$ 2.8***	
DEA	5	95	8.5 $\pm$ 0.7	–	5	98	9.5 $\pm$ 0.7	–	5	100	10 $\pm$ 1.4	–
	10	85	9 $\pm$ 1.4		10	92	9 $\pm$ 2.8		10	95	10.5 $\pm$ 0.7	
	50	80	13.5 $\pm$ 2.1		50	80	12.5 $\pm$ 0.7		50	88	10.5 $\pm$ 2.1	
	100	74	17 $\pm$ 1.4		100	65	14 $\pm$ 1.4		100	60	11 $\pm$ 1.4	

Doses: Pesticide concentrations.

CBPI: Cytokinesis Blocked Proliferative Index.

BMC: Bi Micronucleated Cells.

MCC: Minimal Clastogenic Concentration.

* $P \leq 0.05$.

** $P \leq 0.01$.

*** $P \leq 0.001$.

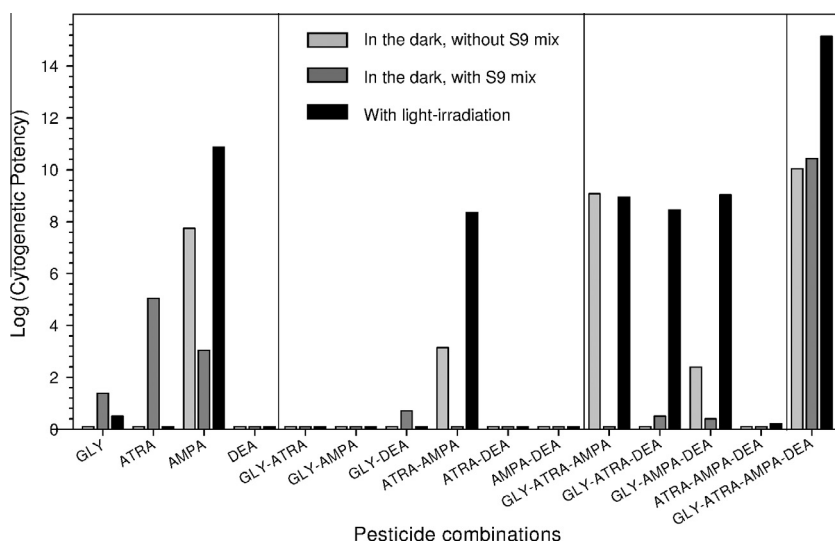


Fig. 1. Summary histogram of the potencies observed for pesticides combinations.

4. Discussion

Results observed in the present study confirmed that glyphosate, atrazine and AMPA exerted cytogenetic effects in mammalian cells. They established that mixtures of pesticides could represent an enhanced genotoxic risk in environmental matrices. They demonstrated moreover that environmental conditions such as sunlight irradiation could greatly modulate the clastogenic/aneugenic profiles of pesticides.

Glyphosate is a broad-spectrum organophosphorus herbicide (Mladinic et al., 2009; Mañas et al., 2009a). AMPA is the main environmental breakdown product of glyphosate. Formed by the metabolic conversion of glyphosate in mammal liver and plants,

its environmental persistence is important, with a half-life period of 76–240 d in soil (Battaglin et al., 2005). As a consequence, AMPA is the most systematically quantified pesticide in French water-courses (Dubois and Lacouture, 2011). Although several regulatory agencies have highlighted the absence of any carcinogenic effect of glyphosate and AMPA in rodent and human (EPA, 1993; FAO/WHO, 1997; E.U., 2002), recent studies have indicated that glyphosate could induce dysfunctions in cell cycle and apoptosis (Marc et al., 2004), endocrine disruption and teratogenicity (Benachour and Séralini, 2009; Paganelli et al., 2010), genetic alterations (Paz-y-Miño et al., 2007) and increase of cancer incidence (De Roos et al., 2005). It has been shown to be non mutagenic in *Salmonella typhimurium*, *Escherichia coli* and *Bacillus subtilis* (Li and Long,

Table 2

Summary data concerning the cytogenetic effects of mixtures of 2 pesticides.

	In the dark				With S9 mix				With light-irradiation			
	Doses ($\mu\text{g mL}^{-1}$)	CBPI (%)	BMC (%e)	MCC ($\mu\text{g mL}^{-1}$)	Doses ($\mu\text{g mL}^{-1}$)	CBPI (%)	BMC (%e)	MCC ($\mu\text{g mL}^{-1}$)	Doses ($\mu\text{g mL}^{-1}$)	CBPI (%)	BMC (%e)	MCC ($\mu\text{g mL}^{-1}$)
Negative control	0	100	10.5 $\pm$ 0.7		0	100			0	100	11.5 $\pm$ 0.7	
Positive control	0.05	99	31 $\pm$ 2.8***		1	98	22 $\pm$ 1.4**		0.05	99	38 $\pm$ 1.4***	
Glyphosate/ atrazine	5	100	8.5 $\pm$ 0.7	–	5	98	10 $\pm$ 1.4	–	5	100	12.5 $\pm$ 0.7	–
	10	98	9.5 $\pm$ 2.1		10	95	11 $\pm$ 2.8		10	97	10.5 $\pm$ 2.1	
	50	91	9 $\pm$ 1.4		50	92	12 $\pm$ 1.4		50	84	10 $\pm$ 1.4	
	100	88	12 $\pm$ 1.4		100	90	12.5 $\pm$ 2.1		100	72	11.5 $\pm$ 2.1	
Glyphosate/AMPA	5	99	10.5 $\pm$ 2.1	–	5	100	10.5 $\pm$ 0.7	–	1	100	9 $\pm$ 1.4	–
	10	98	9.5 $\pm$ 2.1		10	98	9.5 $\pm$ 2.1		150	95	11.5 $\pm$ 0.7	
	50	96	13 $\pm$ 1.4		50	98	11 $\pm$ 1.4		10	88	11 $\pm$ 1.4	
	100	94	15.5 $\pm$ 0.7		100	95	15.5 $\pm$ 2.1		50	68	11.5 $\pm$ 2.8	
Glyphosate/DEA	5	100	9.5 $\pm$ 2.1	–	5	99	12.5 $\pm$ 2.1	22.1	5	100	11.5 $\pm$ 0.7	–
	10	98	9.5 $\pm$ 0.7		10	97	19 $\pm$ 2.8		10	95	12 $\pm$ 2.8	
	50	95	10.5 $\pm$ 0.7		50	93	25 $\pm$ 1.4 **		50	83	9 $\pm$ 1.4	
	100	90	16.5 $\pm$ 2.1		100	90	34 $\pm$ 1.4***		100	75	11 $\pm$ 2.8	
Atrazine/AMPA	0.05	99	10.5 $\pm$ 0.7	0.39	5	98	11.5 $\pm$ 0.7	–	0.001	100	14 $\pm$ 2.1	0.0026
	0.1	95	14.5 $\pm$ 0.7		10	96	11 $\pm$ 2.8		0.005	97	24.5 $\pm$ 0.7**	
	0.5	92	20 $\pm$ 1.4*		50	92	13.5 $\pm$ 0.7		0.01	92	28.5 $\pm$ 2.1***	
	1	87	30 $\pm$ 3.5***		100	89	15 $\pm$ 1.4		0.05	88	33 $\pm$ 1.4***	
Atrazine/DEA	5	100	9.5 $\pm$ 0.7	–	5	98	10.5 $\pm$ 2.8	–	0.5	100	9 $\pm$ 2.8	–
	10	97	11 $\pm$ 1.4		10	92	10 $\pm$ 1.4		1	94	10 $\pm$ 1.4	
	50	93	17 $\pm$ 1.4		50	80	12.5 $\pm$ 0.7		5	86	12.5 $\pm$ 0.7	
	100	86	18.5 $\pm$ 2.1		100	65	11 $\pm$ 1.4		10	72	16 $\pm$ 1.4	
AMPA/DEA	5	95	10.5 $\pm$ 0.7	–	5	99	9.5 $\pm$ 0.7	–	5	99	10 $\pm$ 1.4	–
	10	85	11 $\pm$ 1.4		10	98	10 $\pm$ 3.5		10	92	9.5 $\pm$ 2.1	
	50	80	13.5 $\pm$ 0.7		50	94	11 $\pm$ 1.4		50	84	10.5 $\pm$ 2.1	
	100	74	16 $\pm$ 1.4		100	90	12.5 $\pm$ 2.1		100	68	12 $\pm$ 1.4	

Doses: Sum of the pesticide concentrations (50% of each pesticide).

CBPI: Cytokinesis Blocked Proliferative Index.

BMC: Bi Micronucleated Cells.

MCC: Minimal Clastogenic concentration.

* $P \leq 0.05$.** $P \leq 0.01$.*** $P \leq 0.001$.

1988). However, it has been able to induce in vitro increases of sister chromatid exchanges, chromosome aberrations (Lioi et al., 1998a; Lioi et al., 1998b), and DNA damages in human cells (Monroy et al., 2005; Mladinic et al., 2009; Mañas et al., 2009a). Moreover, it has been shown genotoxic and clastogenic in vivo (Prasad et al., 2009; Mañas et al., 2009a). Contrary to glyphosate, the genotoxic properties of AMPA have been poorly assessed. However, it has been shown to induce chromosome aberrations in human lymphocytes and DNA-lesions in Hep-2 cells (Mañas et al., 2009b) and to generate micronuclei in vivo (Kier and Stegeman, 1993). Results observed in the present study confirmed the cytogenetic toxicity of both compounds. In the dark, glyphosate was not directly active, but it induced micronuclei in the presence of the metabolic fraction S9 mix. On the contrary, AMPA displayed a direct cytogenetic toxicity which was 1000 fold higher than that of its parent compound. Since AMPA has been shown the main metabolite of glyphosate in mammals, this result suggested that the cytogenetic effects observed for glyphosate with S9 mix were mainly due to its metabolic conversion into AMPA. Thus, the weak mutagenic activity of glyphosate with S9 mix could be explained by its weak metabolization yield as observed in vivo (EPA, 1993; FAO/WHO, 1997; E.U., 2002). After sunlight irradiation, glyphosate was weakly active, whereas the cytogenetic properties of AMPA were 20-fold increased. Since the photo-degradation of glyphosate by UV light has been shown to produce AMPA (Assalin et al., 2010), the photo-inducible genotoxic potential of glyphosate could be partially explained by the formation of its metabolite AMPA in

the intracellular environment. However, in the present study, cells were submitted to a very low irradiation procedure that did not induce intrinsic DNA damage. This irradiation was far lower than that used by Manassero et al. (2010), who obtained a 50% degradation of glyphosate after a 5 h exposure to UV irradiation. Results observed in the present study implied more probably the involvement of a photo-induced oxidative stress, which transformed glyphosate and AMPA into photo-activated intermediates and favored their interactions with cellular targets. Glyphosate has been demonstrated to induce changes in the cell redox state, with the production of reactive oxygen species (Bagchi et al., 1995; Lioi et al., 1998b). This production has been linked to the deregulation of the antioxidant defense system (Lioi et al., 1998b; Pieniazek et al., 2004; Beuret et al., 2005; Mañas et al., 2009a; Gehin et al., 2005), which could be enhanced in the presence of light.

Atrazine is a systemic triazine herbicide that inhibits Photosystem II. Since the 1950s, it has been one of the most widely used agricultural herbicides in the world (Cavas, 2011). Although the use of atrazine has been restricted or banned in several countries since the 1990s, it is one of the most frequently detected pesticides in agricultural streams, rivers and groundwaters (Zeljczic et al., 2006; Vryzas et al., 2011; Cavas, 2011; Dubois and Lacouture, 2011). DEA is the major degradation product of atrazine. Formed by hydrolysis or microbial conversion of the parent molecule, it is primarily responsible for groundwater contamination in France (Dubois and Lacouture, 2011). Although atrazine has been classified as “possibly carcinogenic to human” (Group 2B) by the

Table 3

Summary data concerning the cytogenetic effects of mixtures with three and four pesticides.

	In the dark				With S9 mix				With light-irradiation			
	Doses ($\mu\text{g mL}^{-1}$)	CBPI (%)	BMC (%)	MCC ($\mu\text{g mL}^{-1}$)	Doses ($\mu\text{g mL}^{-1}$)	CBPI (%)	BMC (%)	MCC ($\mu\text{g mL}^{-1}$)	Doses ($\mu\text{g mL}^{-1}$)	CBPI (%)	BMC (%)	MCC ($\mu\text{g mL}^{-1}$)
Negative control	0	100	10.5 $\pm$ 2.1		0	100	10.5 $\pm$ 0.7		0	100	12.5 $\pm$ 2.1	
Positive control	0.05	98	31.5 $\pm$ 0.7***		1	99	24 $\pm$ 2.8**		0.05	97	34.5 $\pm$ 3.5***	
Glyphosate/ atrazine/AMPA	0.0005	100	10.5 $\pm$ 0.7	0.001	5	100	10 $\pm$ 2.8	–	0.0005	99	14 $\pm$ 2.8	0.009
	0.001	98	18 $\pm$ 1.4		10	99	12.5 $\pm$ 0.7		0.001	91	20 $\pm$ 2.8*	
	0.005	97	24.5 $\pm$ 3.5**		50	90	11.5 $\pm$ 3.5		0.005	85	28.5 $\pm$ 2.1***	
	0.01	92	26 $\pm$ 2.8***		100	82	15 $\pm$ 1.4		0.01	80	32 $\pm$ 1.4***	
Glyphosate/ atrazine /DEA	5	100	9.5 $\pm$ 3.5	–	5	99	10.5 $\pm$ 0.7	36	0.0005	100	15.5 $\pm$ 0.7	0.001
	10	98	14 $\pm$ 1.4		10	98	14 $\pm$ 1.4		0.001	96	20 $\pm$ 1.4	
	50	95	19.5 $\pm$ 2.1		50	90	20.5 $\pm$ 2.1*		0.005	87	26 $\pm$ 2.8**	
	100	90	22.5 $\pm$ 0.7**		100	84	24.5 $\pm$ 3.5**		0.01	85	33.5 $\pm$ 2.1***	
Glyphosate/AMPA/ DEA	0.5	100	14.5 $\pm$ 0.7	0.82	5	100	12 $\pm$ 1.4	46	0.0005	100	14 $\pm$ 2.8	0.0008
	1	99	22.5 $\pm$ 2.1**		10	100	11.5 $\pm$ 2.1		0.001	98	22.5 $\pm$ 0.7*	
	5	96	30 $\pm$ 1.4***		50	98	19.5 $\pm$ 3.5		0.005	93	33 $\pm$ 2.8**	
	10	92	35 $\pm$ 2.8***		100	86	28 $\pm$ 2.8***		0.01	85	40 $\pm$ 1.4***	
Atrazine/AMPA/ DEA	5	100	10 $\pm$ 1.4	–	5	99	11.5 $\pm$ 2.1	–	5	100	9.5 $\pm$ 2.1	74
	10	96	11.5 $\pm$ 2.1		10	98	10 $\pm$ 1.4		10	94	10.5 $\pm$ 0.7	
	50	92	14 $\pm$ 1.4		50	91	9.5 $\pm$ 2.1		50	82	15.5 $\pm$ 2.1	
	100	88	18 $\pm$ 2.8		100	85	12.5 $\pm$ 0.7		100	65	22 $\pm$ 2.8**	
Glyphosate/ Atrazine/AMPA/ DEA	0.0001	99	14 $\pm$ 2.8	0.0004	0.0001	100	9.5 $\pm$ 2.1	0.0003	1 $\times$ 10 ^{–6}	99	15 $\pm$ 1.4	4 $\times$ 10 ^{–6}
	0.0005	100	23 $\pm$ 2.8**		0.0005	99	20.5 $\pm$ 2.1**		5 $\times$ 10 ^{–6}	99	23.5 $\pm$ 2.1**	
	0.001	98	30.5 $\pm$ 3.5***		0.001	99	34 $\pm$ 1.4***		1 $\times$ 10 ^{–5}	98	31 $\pm$ 2.8***	
	0.005	97	42 $\pm$ 2.8***		0.005	98	44 $\pm$ 2.8***		5 $\times$ 10 ^{–5}	97	38 $\pm$ 2.8***	

Doses: Sum of the pesticide concentrations (33.33% of each pesticide for mixture with 3 pesticides, 25% of each pesticide for mixture with 4 pesticides).

CBPI: Cytokinesis Blocked Proliferative Index.

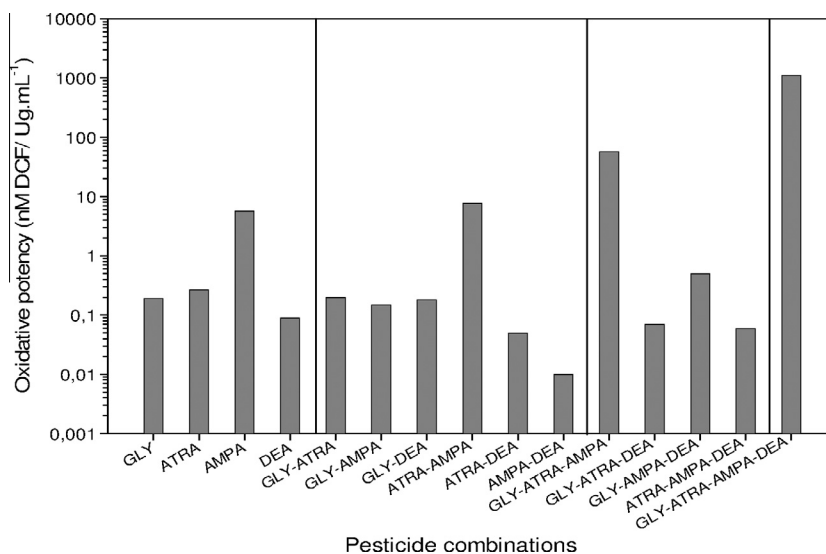
BMC: Bi Micronucleated Cells.

MCC: Minimal Clastogenic concentration.

* $P \leq 0.05$.** $P \leq 0.01$.*** $P \leq 0.001$.

International Agency for Research on Cancer (1999), the US Environmental Protection Agency has classified this pesticide as “not likely to be a human carcinogen (group D)” in 2003. Data concerning the in vitro genotoxicity of atrazine are rather conflicting and mainly dependent on experimental conditions. Atrazine has been shown deprived from mutagenicity in bacteria and fungi (Ruiz and Marzin, 1997). However, it has been shown to induce chromosome damages in CHO cells (Biradar and Rayburn, 1995; Taets et al., 1998; Rayburn et al., 2001) and in human lymphocytes

(Meisner et al., 1992; Ribas et al., 1995; Ribas et al., 1998; Lioi et al., 1998b; Malik et al., 2004), and to generate micronuclei in vivo (Gebel et al., 1997). Results observed in the dark confirmed these data, since they demonstrated that atrazine induced micronuclei in the presence of the metabolic fraction S9 mix. They were in agreement with those of Lioi et al. (1998b), who suggested that the clastogenic mechanism of atrazine was mediated by the induction of a pro-oxydant state of the cells, with a significant enhancement of the glucose 6-phosphate dehydrogenase activity. In

**Fig. 2.** Summary histogram of the intracellular ROS amount observed for pesticides combinations.

mammalian liver, atrazine is mainly transformed into bidealkylatrazine and, in a lower extend, into deisopropylatrazine and DEA. The observation that DEA was inactive in CHO cells in all the experimental conditions indicated that bidealkylatrazine and/or deisopropylatrazine were responsible for the mutagenic activity observed for atrazine in the presence of S9 mix. No photo-inducible activity could be observed with atrazine and DEA. This result was in accordance with previously published data, since a low photo-inducible toxicity has been shown on marine bacteria and freshwater mesocosms (Lin et al., 1999; Detenbeck et al., 1996; Pelletier et al., 2006).

The genotoxic properties of pesticide mixtures as compared to those of individual compounds have been poorly investigated. However, in environmental compartments, a large variety of pesticides including glyphosate, atrazine and their degradation products AMPA and DEA, as well as many other active compounds such as polycyclic aromatic hydrocarbons and metals (White and Claxton, 2004), have been currently detected. In the present study, various pesticide mixtures were made with two, three or the four pesticides, and assessed for their cytogenetic effects. Results revealed that the mutagenic potentials of pesticide mixtures greatly varied according to their chemical composition and to the experimental conditions. Surprisingly, all the mixtures made with two pesticides except the atrazine/AMPA mixture, were weakly genotoxic whatever the experimental conditions considered. AMPA appeared to be inactivated when combined with glyphosate or DEA, but it remained strongly genotoxic when combined with atrazine. Similarly, all the mixtures made with three pesticides except the glyphosate/atrazine/AMPA mixture, exerted a low cytogenetic toxicity in the dark. They were all inactivated by the metabolic fraction, and all except the atrazine/AMPA/DEA mixture, exhibited a high cytogenetic toxicity after light irradiation. These results established the strong cytogenetic potential of AMPA when combined with atrazine and glyphosate, and its inactivation when combined with atrazine and DEA. The mixture made with the four pesticides exhibited a more potent cytogenetic toxicity than those of individual pesticides in all the experimental conditions. It was 20-fold higher than those of AMPA in the dark, and it was 100-fold enhanced as compared to those of AMPA after light irradiation. These results suggested that addition of glyphosate in the non-genotoxic atrazine/AMPA/DEA mixture was determinant for the genotoxic potential.

Oxidative stress, probably due to unstable chemical interactions between the different pesticides, has been hypothesized to explain the genotoxic activity of pesticides (Banerjee et al., 2001). Results observed in the present study showed that the most genotoxic pesticide mixtures induced the most important oxidative stress in the intracellular environment. They suggested that oxidative stress could play an important role in the deleterious consequences of pesticide mixtures, as already demonstrated by Coleman et al. (2012).

Synergistic activities have already been observed with roundup, a commercial mixture of glyphosate and the polyethoxylated tallow amine surfactant, and with other pesticides (Koller et al., 2012; Kocaman and Topaktaş, 2010; Graillot et al., 2012). They indicated that the classical toxicology studies, based on the testing of individual molecules, were not efficient to estimate the global environmental risk. As an example, the maximal regulatory concentration has been set at $0.1 \mu\text{g L}^{-1}$ for each pesticide in drinking waters in France (Directive of the Council of European Communities CEE No. 80-778 of 15/07/80 on the quality of water intended for human consumption). At this concentration, none of the pesticides or pesticide metabolites was clastogenic. When considering pesticide mixtures, the maximal regulatory concentration has been set at $0.5 \mu\text{g L}^{-1}$ for the sum of the different compounds. Results observed in the present study demonstrated that this concentra-

tion was clastogenic/aneugenic with the glyphosate/atrazine/AMPA/DEA mixture in all the experimental conditions ($\text{MCC} < 0.4 \mu\text{g L}^{-1}$). In European rivers, glyphosate concentrations have been shown to reach $75\text{--}90 \mu\text{g L}^{-1}$ after rainfall events, while glyphosate, atrazine, atrazine-desethyl and AMPA concentrations have been shown to usually exceed the European quality standard concentrations of $0.1 \mu\text{g L}^{-1}$ in drinking waters (Botta et al., 2009; Migeot et al., 2013). These concentrations have been shown weakly toxic on aquatic biota (Katagi, 2010), but their genotoxic impact that occurs at much lower concentrations has been poorly evaluated.

5. Conclusion

Results observed in the present study clearly established that enhanced cytogenetic activities due to cocktail effects could be observed in pesticide mixtures containing glyphosate, atrazine, and their degradation products AMPA and DEA. They also demonstrated that light-irradiation, corresponding to a few minutes of solar exposure, greatly potentiated this cytogenetic impact. Due to the persistence of the widely used pesticides glyphosate and atrazine in soils, such mixtures are certainly present in several environmental compartments. As a consequence, they represent an additional genotoxic risk for ecosystems and human health.

Today, the ability of pesticides to cause DNA damage represents an important issue for their approval, and toxicology studies, mainly based on the testing of individual molecules on rodents, have been developed to predict their genotoxic potential in human. Experiments performed on cultured cells, that imperfectly mimic whole animals, are generally considered as preliminary assays. Nevertheless, they allow testing a great number of molecules or molecule mixtures at various concentrations, in various environmental conditions. In the present study, in vitro experiments showed that the genotoxic impact of pesticides greatly depended on their physico-chemical environment, and that pesticide mixtures could reveal genotoxic properties at concentrations which were far lower than those of the individual molecules. As a consequence, they pointed out the limits of usual testing strategies to efficiently estimate environmental risks, and suggested that the Directive Standards for Pesticides in Drinking Water should be re-evaluated according to these under-estimated factors of risk (Dolan et al., 2013).

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